Fiscal Year:	FY 2007	**Task Last Updated:**	FY 06/29/2007
PI Name:	Cavanagh, Peter R. Ph.D., D.Sc.		
Project Title:	Bone Strain During 0G Exercise		
Division Name:	Human Research		
Program/Discipline:	HUMAN RESEARCH		
Program/Discipline--Element/Subdiscipline:	HUMAN RESEARCH--Operational and clinical research		
Joint Agency Name:		**TechPort:**	No
Human Research Program Elements:	None		
Human Research Program Risks:	None		
Space Biology Element:	None		
Space Biology Cross-Element Discipline:	None		
Space Biology Special Category:	None		
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City:	Seattle	**State:**	WA
Zip Code:	98195-6500	**Congressional District:**	7
Comments:	PI moved from Cleveland Clinic to University of Washington in June 2008 (8/08)		
Project Type:	Ground	**Solicitation / Funding Source:**	99-HEDS-03
Start Date:	11/01/2003	**End Date:**	12/31/2007
No. of Post Docs:	0	**No. of PhD Degrees:**	0
No. of PhD Candidates:	0	**No. of Master' Degrees:**	0
No. of Master's Candidates:	0	**No. of Bachelor's Degrees:**	0
No. of Bachelor's Candidates:	0	**Monitoring Center:**	NASA JSC
Contact Monitor:		**Contact Phone:**	
Contact Email:			
Flight Program:			
Flight Assignment:	No cost extension through 12/31/07 (per PI 5/07) No cost extension through 12/31/06 (per J. Dardano 2/06)		
Key Personnel Changes/Previous PI:	n/a		
COI Name (Institution):			
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	Considerable NASA resources have been devoted to a vibration-isolated treadmill (TVIS) that has been installed in the International Space Station (ISS). This device is attractive because programs may be designed to allow exercise countermeasures to be applied against bone, muscle, and cardiovascular adaptations. Despite the need for on-orbit exercise using TVIS, almost no research has been conducted to determine the acceptability of the subject load device (SLD) and harness used to tether crewmembers to the treadmill. Data regarding the bone strains produced by exercise countermeasures, such as running in reduced gravity, are also unavailable, yet there is little doubt that bone strain plays an important intermediary role in the signal transduction pathway responsible for bone maintenance and functional adaptation. This study in simulated microgravity was designed to evaluate the comfort and function of a new SLD design applied to subjects running in a system that allows controlled distribution of the load between the waist and shoulders during treadmill running. Using the Zero Gravity Locomotion Simulator (ZLS), twelve subjects completed nine, five minute running trials at a constant speed (3.35 ms-1) using 3 SLD loads (50%, 75% and 100% BW) presented at random while using a 50:50 shoulder-to-waist loading ratio. Subjective ratings of discomfort, ground reaction forces and SLD loads were collected. 100% BW in the ZLS resulted in similar GRF profiles to those obtained from the literature for overground running. The full body weight loading conditions were comfortably tolerated. The ZLS and its SLD system are promising methods of simulating overground running in zero gravity as evidenced by the low discomfort levels of subjects and their ground reaction force profiles. A new pneumatic SLD system has been implemented increase comfort and reduce load variation. A new harness utilizing backpack technology has also been constructed to improve subject comfort while running on the ZLS.
Task Description:	The second phase of this investigation is examining the feasibility of the new system in a long-term exercise program, such as the 3 to 6 months that crewmembers are currently spending on the ISS. Subjects are running four times per week on the ZLS for three continuous weeks. During each session, the subjects run for 30 minutes total at a pace of 2.68 - 3.49 ms-1 while loaded to 100% BW. During the third phase, data collected on the ZLS will then be used as input to robotic cadaver simulations in order to directly measure bone strains in cadaver limbs moving through accurate kinematic and kinetic reconstructions of orbit-based locomotion. Thus, these sequential experiments will provide the unique psychophysical and biomechanical information needed to design better exercise countermeasures to bone loss on the ISS. They will also provide fundamental new data on the relationship between external loading and strain in human bone. The specific aims of this project are: 1. To define the parameters (load, load distribution, and exercise frequency) of a treadmill exercise program using the SLD harness system that would be acceptable to the crew during prolonged space flight. 2. To measure gravity replacement loads, kinematics, ground reaction forces, muscle activity, and subjective ratings of comfort during the exercises defined in specific aim #1. 3. To use data generated in specific aim #2 to extend the capabilities of a parallel robot (Rotopod R2000) that acts as a cadaveric dynamic gait simulator, which will enable measurement of bone strains generated during on-orbit treadmill running. 4. To determine bone strain at several sites on the distal tibia during walking in 1g, and during the simulated on-orbit conditions identified in specific aim #1.
Rationale for HRP Directed Research:	
Research Impact/Earth Benefits:	The research aim of this study is to provide psychophysiological and biomechanical information to design better exercise programs to prevent bone loss.
Task Progress:	NASA Glenn Research Center and the Cleveland Clinic (CCF) collaborated to create a passive pneumatic subject load device (pSLD). This device uses a single pneumatic cylinder to provide a nearly constant spring force on the cables attached to the subject harness. A new SLD harness, which leverages existing backpack technology, which includes a cupped and canted waist belt and “S”-shaped shoulder straps, has been constructed. This harness allows subjects to self-regulate the distribution of the load between the shoulders and hips by tightening or loosening the harness shoulder straps. Experiment #1 has demonstrated that the application of 100% BW loads to a subject running in simulated micro-gravity can generate GRF’s similar to normal 1-g running. The harness discomfort ratings collected indicate that running under these conditions is tolerable for short durations. As expected, harness discomfort increased in all the rated body regions as the SLD load was increased from 75% BW to 100% BW. The harness discomfort during longer duration runs while loaded to 100% BW in simulated microgravity are currently being evaluated as part of experiment #2. Discomfort ratings for the lower extremity and the body regions affected by the harness are being collected and will be analyzed longitudinally. The hip to shoulder load ratio is being captured by a pair of force transducers in order to quantify the variability of the self-regulated ratio between subjects. One subject has successfully completed the entire protocol. Two other subjects have participated in the experiment but withdrew before completion. One withdrew due to an injury unrelated to the study, and the other withdrew due to personal reasons. One subject is currently participating in the study, and several additional subjects have begun the screening process for participation. A total of 8 subjects are anticipated to have completed the protocol by November. Considerable progress has been made on the musculoskeletal simulator which will be utilized as part of experiment #3. A force platform has been mounted onto the Rotopod, a rotary tendon actuator integrated into the system, and an operator interface developed. The bone strain data acquisition system has also been developed and demonstrated. Synchronization between the Rotopod and tendon actuator has been verified via simulated walking, which fully loaded the forefoot of a cadaver foot via actuation of the Achilles’ tendon. The optimization algorithm which adjusts the Rotopod trajectory and tendon force application is currently being verified. Once completed, five additional rotary tendon actuators will be integrated into the musculoskeletal simulator.
Bibliography Type:	Description: (Last Updated: 03/08/2018)
Articles in Peer-reviewed Journals	Genc KO, Mandes VE, Cavanagh PR. "Gravity replacement during running in simulated microgravity." Aviat Space Environ Med. 2006 Nov;77(11):1117-24. PMID: 17086763 , Nov-2006